UNITED STATES PATENT AND TRADEMARK OFFICE
PROVISIONAL PATENT APPLICATION UNDER 35 U.S.C. § 111(B)

TITLE OF THE INVENTION: SYSTEM AND METHOD FOR OPTICAL MENU PRICE-ARBITRAGE ANALYSIS, TWO-TIER CHEMICAL SULFITE FALLBACK INFERENCE, AND CROSS-MODAL SENSORY PALATE CALIBRATION
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CROSS-REFERENCE AND FIELD OF THE INVENTION

[0001] The present invention relates generally to computer vision, optical character recognition (OCR), automated chemical and allergen concentration inference, retail price-arbitrage calculation engines, and personalized recommender systems. More specifically, the invention relates to a computer-implemented system and method that ingests unstructured restaurant beverage menus (via optical camera capture or venue-level inventory resolution) and simultaneously computes: (1) real-time on-premise menu-to-off-premise retail markup arbitrage and Quality-to-Price Ratio (QPR); (2) two-tier chemical sulfite (SO2) concentration and physiological reaction risk via an automated fallback inference engine; and (3) cross-modal sensory palate compatibility derived from non-wine bitterness/acidity proxy inputs and longitudinal user cellar ratings.

BACKGROUND OF THE INVENTION

[0002] Consumers ordering wine in on-premise hospitality environments (restaurants, steakhouses, wine bars) face three compounding informational asymmetries. First, restaurants apply highly non-uniform price markups across their beverage lists—frequently marking up recognizable mass-market by-the-glass and entry-level bottles by 300% to 400% (3.0x to 4.0x off-premise retail price) while marking up certain allocation or estate bottles by only 135% to 160% (1.35x to 1.60x retail). Existing beverage applications require a physical bottle label or only display crowd-sourced ratings without parsing multi-column glass-versus-bottle menu pricing or computing the venue-specific markup arbitrage ratio.

[0003] Second, millions of consumers experience adverse physiological reactions (vascular headaches, flushing, nasal congestion) triggered by elevated free and total sulfur dioxide (SO2), histamines, or synthetic fining agents in conventional winemaking (150 to 350 ppm). However, mainstream restaurant menus rarely carry strictly certified "No Sulfites Added" (NSA, <10 ppm) labels. Conventional filtering systems employ binary boolean filters that return zero results on 90% of restaurant menus when no strictly sulfite-free SKU is present.

[0004] Third, consumers frequently cannot articulate technical enological parameters (e.g., malolactic conversion, phenolic tannin structure), causing standard filter-based applications to recommend mismatched bottles.

SUMMARY OF THE INVENTION

[0005] The present invention provides a 4-Vector Menu Intelligence Engine comprising: (a) a Dual-Price Optical Menu Parser that disambiguates paired numerical tokens into on-premise by-the-glass price (P_glass) and on-premise bottle price (P_bottle), synthesizing implied bottle pricing when only glass pricing is printed (P_bottle = 4 * P_glass); (b) a Real-Time Retail Markup Arbitrage and QPR Engine that computes a venue-specific Markup Multiple (M_i = P_bottle,i / P_retail,i), dollar savings relative to industry mean markup (2.75x), and a composite QPR score; (c) a Two-Tier Sulfite (SO2) Fallback Inference Engine that isolates Tier 1 (Strictly No Added Sulfites / NSA, 0 ppm added SO2) SKUs first, and automatically triggers a Tier 2 Fallback Protocol (identifying Certified Organic, Demeter Biodynamic, dry-farmed, native yeast, and clay amphora markers with ~18-45 ppm estimated SO2) when Tier 1 inventory is absent; and (d) a Cross-Modal Sensory Proxy and Collaborative Cellar Loop that translates non-wine sensory preferences alongside longitudinal user star ratings (1 to 5 stars) into a real-time Personal Match Score (P_u,i).

DRAWINGS AND SYSTEM ARCHITECTURE

FIG. 1 — 4-VECTOR MENU INTELLIGENCE & ARBITRAGE SYSTEM ARCHITECTURE

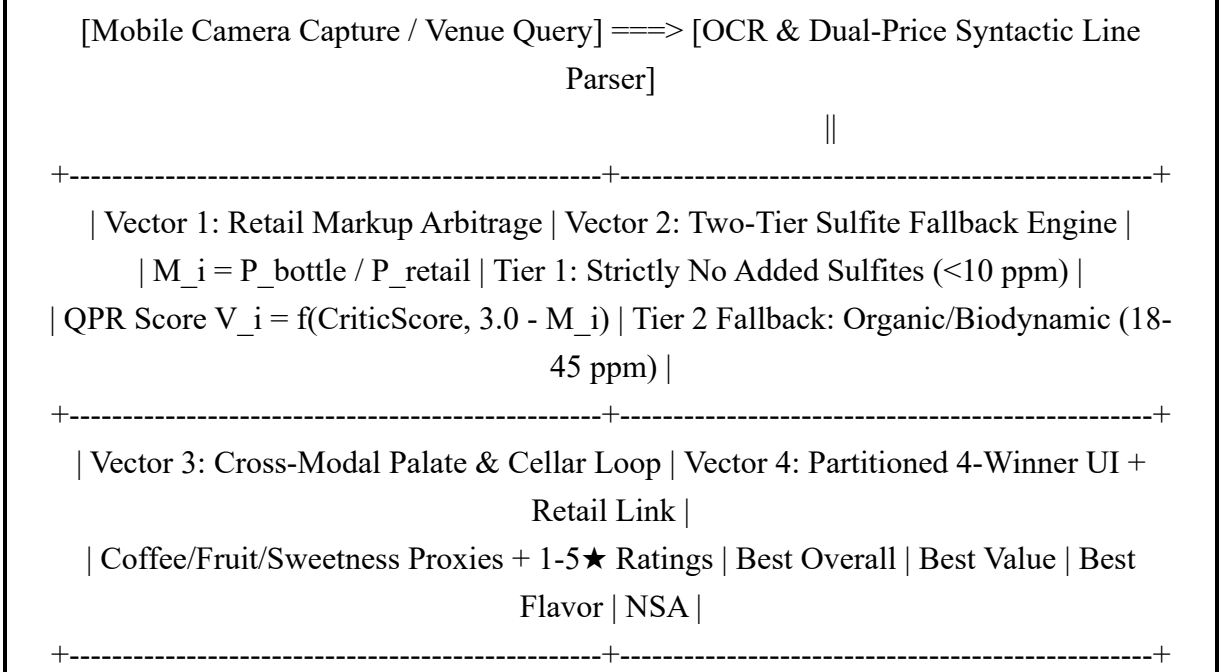
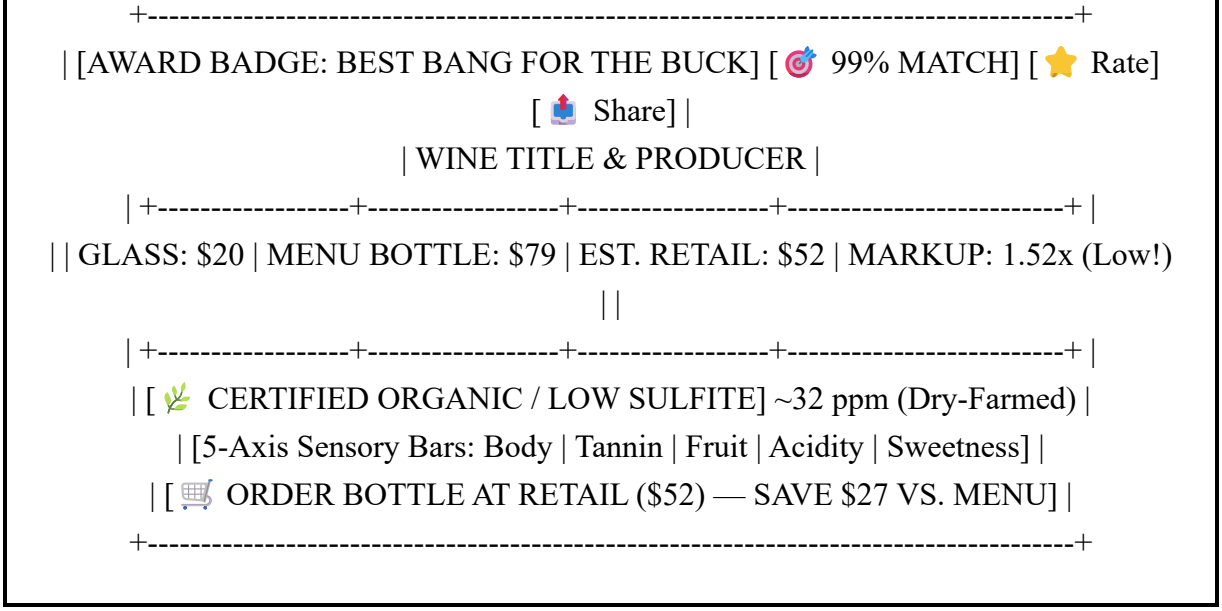


FIG. 2 — ORNAMENTAL UI CARD LAYOUT (DESIGN CLAIM)



DETAILED MATHEMATICAL FORMULATIONS

[0006] For each candidate wine line item i extracted from an optical menu image, the parser identifies numeric price tokens {p_1, p_2} in [10, 600]. When p_1 < 40 and p_2 >= 35, the system assigns P_glass,i = p_1 and P_bottle,i = p_2. The Restaurant Markup Multiple M_i and Value Score V_i are computed as:

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M_i = P_bottle,i / max(P_retail,i, 1)
Savings_i = max(0, round((2.75 - M_i) * P_retail,i))
V_i = min(99, round(1.15 * (Q_i - 50) + max(0, 18 * (3.0 - M_i)) +
Alpha(P_bottle,i)))
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[0007] Given a user u with cross-modal quiz vector q_u = (q_coffee, q_fruit, q_sweet, q_sulfite, q_budget) and historical cellar rating set R_u = {(w_k, r_u,k)} where r_u,k in {1..5}, the Personal Palate Match Score P_u,i is dynamically adjusted by cellar feedback: if r_u,i >= 4, P_u,i = 99; if r_u,i <= 2, P_u,i = P_u,i - 20; if varietal v_i matches any cellar wine k with r_u,k >= 4, P_u,i = P_u,i + 5.

CLAIMS

1. A computer-implemented menu intelligence and beverage recommendation system comprising one or more processors and a non-transitory computer-readable storage medium storing instructions that, when executed, cause the system to: (a) ingest an unstructured beverage menu dataset associated with a dining venue via optical character recognition (OCR) of a captured menu image or a venue-specific inventory lookup; (b) parse each line item of the unstructured beverage menu dataset to extract beverage identifiers, varietal classifications, and dual on-premise price tokens comprising a by-the-glass price and a by-the-bottle price; (c) calculate, for each parsed beverage item, an on-premise-to-off-premise retail markup multiple (M_i) and a composite Quality-to-Price Ratio (QPR) score by comparing the extracted on-premise price against an off-premise retail benchmark database; (d) execute a two-tier chemical sulfite (SO2) classification and fallback routine that classifies beverage items having zero added sulfur dioxide into a first tier (Tier 1 NSA), classifies beverage items exhibiting organic, biodynamic, dry-farmed, or native-yeast fermentation markers into a second ultra-low-sulfite tier (Tier 2) with an estimated parts-per-million (ppm) sulfite concentration, and automatically surfaces Tier 2 beverage items as primary clean-wine recommendations when a count of Tier 1 beverage items on the ingested menu equals zero; (e) compute a personalized

palate match score (P_u,i) for each parsed beverage item by mapping non-wine sensory proxy responses from a user profile alongside historical user star ratings stored in a personal digital cellar; and (f) render a partitioned multi-winner display simultaneously presenting a Best Overall selection, a Best Bang for the Buck (lowest markup arbitrage) selection, a Best Flavor/Palate Match selection, and a Two-Tier Sulfite-Free/Low-Sulfite selection.

2. The system of Claim 1, wherein the non-wine sensory proxy responses comprise a coffee bitterness/roast preference calibrated to phenolic tannin tolerance, a citrus-versus-dark-fruit flavor preference calibrated to titratable acidity and varietal family, and a physiological headache/sulfite sensitivity indicator.

3. The system of Claim 1, wherein storing a user star rating of 4 or 5 stars for a first beverage item in the personal digital cellar automatically increments the personalized palate match score of subsequent menu items sharing a grape varietal or flavor style with the first beverage item.

4. The system of Claim 1, further comprising a social graph sharing module configured to transmit a structured beverage recommendation payload between user profiles via an in-application social feed and an automated Short Message Service (SMS) deep link containing a sender profile identifier.

5. The system of Claim 1, further comprising an off-premise retail arbitrage transaction module configured to display, adjacent to the on-premise restaurant bottle price, an actionable retail acquisition link enabling the user to purchase the parsed beverage item for off-premise delivery at the lower retail benchmark price.

6. The ornamental design for a mobile display screen with an integrated Four-Column Restaurant-to-Retail Markup Grid, Inline Two-Tier Sulfite PPM Pill, Personal Match Percentage Badge, and Five-Axis Sensory Bar Meter, substantially as shown and described in FIG. 2.

ABSTRACT OF THE DISCLOSURE

A computer-implemented system and method for analyzing restaurant beverage menus via optical character recognition (OCR) and venue inventory resolution. The system extracts paired by-the-glass and by-the-bottle menu prices, computes real-time on-premise-to-retail markup multiples to identify under-marked value bottles ("Best Bang for the Buck"), executes a two-tier chemical sulfite (SO2) classification and fallback inference routine separating strictly no-added-sulfite (NSA) wines from organic/biodynamic low-sulfite alternatives, and

calibrates personalized palate match scores using non-wine sensory proxies and longitudinal user cellar ratings.